

A/B Test Analysis: New Landing Page Evaluation

Portfolio case

Executive Summary

An e-commerce company tested a new landing page against the current page. The goal of the experiment was to determine whether the new page increases user conversion.

The new page did not produce a statistically significant improvement. The control group conversion rate was **12.04%**, while the treatment group conversion rate was **11.88%**. The absolute effect was **-0.158 percentage points**, with a p-value of **0.190**. The 95% confidence interval for the effect was **[-0.394 pp; +0.078 pp]**. Under explicit business assumptions, the observed effect would imply approximately **\$142k lower monthly gross profit** if rolled out.

Recommendation: do not roll out the new landing page in its current form.

1 Business Problem

The experiment answers the following business question:

Does the new landing page increase user conversion compared with the old page?

The primary metric is conversion rate:

$$\text{Conversion Rate} = \frac{\text{Converted Users}}{\text{Total Users}}$$

A positive and statistically significant effect would support a full rollout of the new page. If the effect is not significant, the available evidence is not strong enough to replace the old page.

2 Dataset

The raw dataset contains **294,478** rows and 5 fields: user identifier, timestamp, experiment group, landing page shown, and binary conversion flag. The experiment ran from **2017-01-02** to **2017-01-24**.

Field	Description
user_id	Unique user identifier
timestamp	Time of experiment exposure
group	Experiment group: control or treatment
landing_page	Page shown to the user: old_page or new_page
converted	Binary conversion flag: 0 or 1

Table 1: Raw dataset structure

3 Data Cleaning

Before estimating the effect, invalid rows were removed:

- control users should see only `old_page`;
- treatment users should see only `new_page`;
- each user should be counted once.

Metric	Value
Raw rows	294,478
Clean user-level rows	290,584
Removed invalid or duplicate rows	3,894

Table 2: Data cleaning result

4 Methodology

A two-proportion z-test was used to compare conversion rates.

Null hypothesis:

$$H_0 : p_{treatment} = p_{control}$$

Alternative hypothesis:

$$H_1 : p_{treatment} \neq p_{control}$$

The treatment effect was calculated as:

$$Lift = p_{treatment} - p_{control}$$

Significance level: $\alpha = 0.05$.

5 Results

Group	Users	Conversions	Conversion Rate
Control: old page	145,274	17,489	12.04%
Treatment: new page	145,310	17,264	11.88%

Table 3: Conversion by experiment group

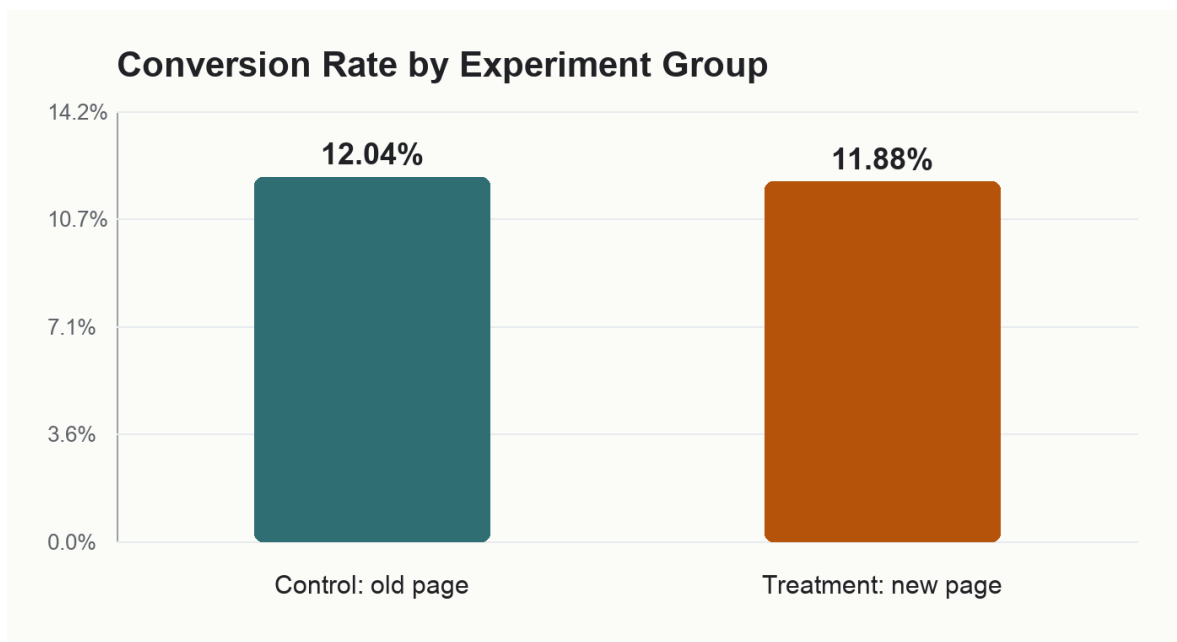


Figure 1: Conversion rate by experiment group

Metric	Value
Absolute lift	-0.158 pp
Relative lift	-1.31%
z-score	-1.311
p-value	0.190
95% confidence interval	[-0.394 pp; +0.078 pp]

Table 4: Final statistical metrics

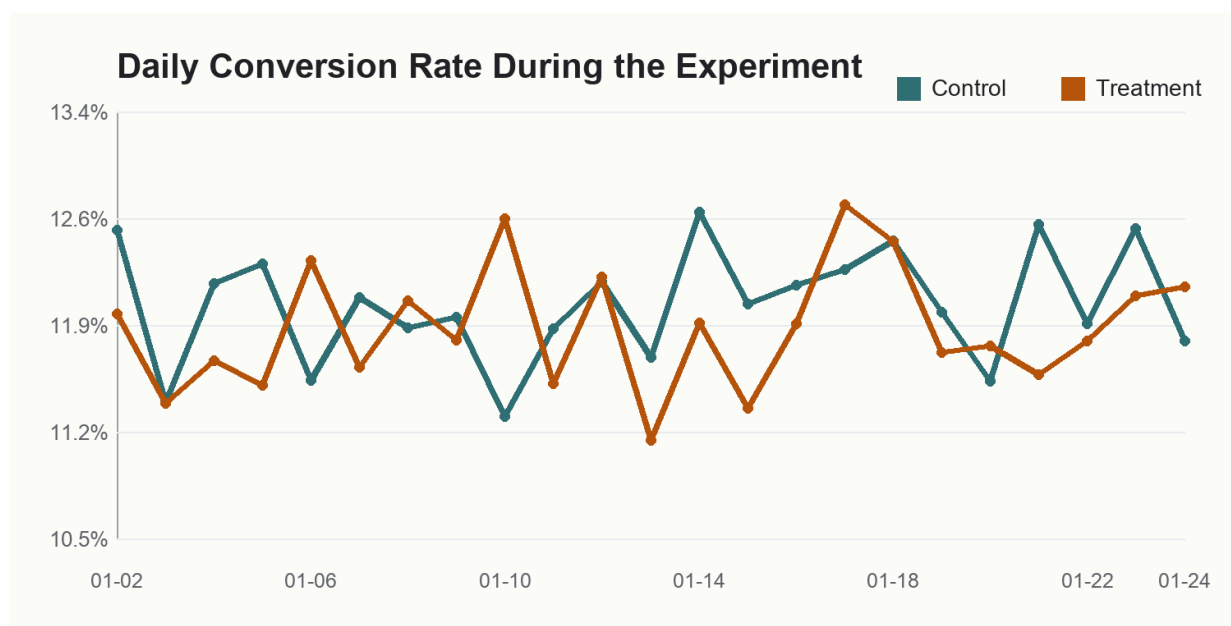


Figure 2: Daily conversion rate trend

6 Interpretation

The treatment group had a slightly lower conversion rate than the control group: **11.88%** versus **12.04%**. However, the p-value was **0.190**, which is above the 0.05 significance threshold. The 95% confidence interval also includes zero.

This means the observed difference may be explained by statistical noise. Based on this experiment, there is not enough evidence to claim that the new landing page improves conversion.

7 Decision Risk and Business Impact

The statistical result was translated into business impact using explicit assumptions. These assumptions are illustrative and should be replaced with company-specific economics before a production decision.

Assumption	Value
Daily users after rollout	100,000
Analysis horizon	30 days
Revenue per conversion	\$75
Gross margin	40%
Significance level	5%
Target power for next test	80%

Table 5: Business impact and planning assumptions

Scenario	Lift	Monthly gross profit impact
Observed effect persists	-0.158 pp	-\$142,041
95% CI lower bound	-0.394 pp	-\$354,411
95% CI upper bound	+0.078 pp	+\$70,328

Table 6: Estimated rollout impact over a 30-day horizon

The downside case is materially larger than the plausible upside case. This asymmetry strengthens the recommendation not to launch the new page as-is.

Minimum detectable effect	Users per group	Total users
0.25 pp	265,967	531,934
0.50 pp	66,492	132,984
1.00 pp	16,623	33,246

Table 7: Approximate sample size plan for future tests

8 Recommendations

1. Do not roll out the new page to all users in its current form.
2. Keep the old page as the default user experience.
3. Review the new page design and identify likely friction points.
4. Before the next test, define the minimum detectable effect and required sample size upfront.

5. Use business impact thresholds in addition to statistical significance when making the rollout decision.
6. Analyze additional segments if business dimensions are available: device, traffic source, country, and new versus returning users.

9 Tools Used

- Python
- pandas
- manual two-proportion z-test
- LaTeX
- SVG visualizations